Project Interim Report

On

Online Quiz Application

(Using Core Java & OOP Concepts)

RU-100-01-00012

Object Oriented Programming with Java

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Abstract

This project focuses on the design and implementation of an Online Quiz Application using Java and Object-Oriented Programming (OOP) concepts. The main aim of the project is to create an interactive system where users can attempt quizzes and get instant results.

Online quizzes are widely used in educational platforms to test knowledge efficiently. This application allows storing questions, displaying multiple-choice options, taking user input, and evaluating answers. The system calculates the final score based on correct responses.

The project demonstrates the use of OOP principles such as classes, objects, encapsulation, and arrays of objects. It helps in understanding real-world application development using Java.

Introduction

The Online Quiz Application is a console-based system developed using Java. It allows users to attempt multiple-choice questions and receive instant feedback.

Objectives:

- **To create an interactive quiz system**
- **To apply Object-Oriented Programming concepts**
- **To automate quiz evaluation**
- **To reduce manual checking effort**

Features:

- **Multiple-choice questions**
- **Instant result calculation**
- **User-friendly console interface**
- **Score tracking system**
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Scope

This project is suitable for:

- Students for practice quizzes
- Small educational systems

Future scope:

- GUI-based system
- Database integration
- Online multiplayer quiz

System Requirements

Hardware:

- Computer with 4GB RAM

Software:

- Java JDK
- IDE (VS Code / Eclipse)

Methodology

The system works in the following steps:

1. Questions are stored using a class
2. Quiz manager controls the flow
3. Questions are displayed one by one
4. User enters answers

5. Answers are checked using .equals()
6. Score is calculated
7. Final result is displayed

Flowchart

Start → Load Questions → Display Question → Take Input → Check Answer → Update Score → Next Question → Show Result → End

· **ClassName**

Question Class

- Variables:
  - questionText
  - options[]
  - correctAnswer
- Methods:
  - Constructor

QuizManager Class

- Variables:
  - Question[] questions
  - score
- Methods:
  - startQuiz()
  - displayQuestions()
  - checkAnswer()
  - showResult()

Implementation

Classes Used:

1. Question Class

- **Stores question, options, and correct answer**
- **Uses constructor for initialization**

2. QuizManager Class

- **Stores multiple questions in array**
- **Displays questions using loop**
- **Takes user input using Scanner**
- **Checks answers using .equals()**
- **Calculates score**

Q1: What is Java?

a) Programming Language

b) OS

c) Database

d) Browser

Enter your answer: a

Correct!

Q2: What is OOP?

Enter your answer: b

Wrong!

Final Score: ½

Gantt Chart

Task	Duration
Planning	2 days
Design	2 days
Coding	4 days
Testing	2 days

Conclusion

The Online Quiz Application successfully demonstrates the use of Java and Object-Oriented Programming concepts. It provides a simple and effective way to conduct quizzes and evaluate user performance.

References

[1] H. Schildt, *Java: The Complete Reference*, McGraw-Hill, 2018

[2] Oracle, "Java

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